

Week 3 - Day 1

Library Catalog REST API Resource Map

1. Domain and Core Resources

Domain:

Library Catalog API

Core Resources:

- books
- authors

Resources are represented as plural nouns following REST naming conventions

2. Primary Resource: Books

The primary resource in this API is **books**.

Book Endpoints

HTTP Method	Endpoint	Description
GET	<code>/api/v1/books</code>	Retrieve all books
GET	<code>/api/v1/books/{id}</code>	Retrieve a specific book by ID
POST	<code>/api/v1/books</code>	Create a new book
PUT	<code>/api/v1/books/{id}</code>	Update an existing book
DELETE	<code>/api/v1/books/{id}</code>	Delete a books

3. Nested Resource

Relationship

An author can have multiple books, while each book belongs to one author.

Relationship:

Author (1) ----- (*) Books

Nested Endpoint

HTTP Method	Endpoint	Description
GET	<code>/api/v1/authors/{authorId}/books</code>	Retrieve all books written by a specific author

4. HTTP Status Codes

Success and Error Responses

Endpoint	Success Status Code	Error Status Code
GET <code>/api/v1/books</code>	200 OK	404 Not Found (No books found)
GET <code>/api/v1/books/{id}</code>	200 OK	404 Not Found (Book not found)
POST <code>/api/v1/books</code>	201 Created	400 Bad Request
PUT <code>/api/v1/books/{id}</code>	200 OK	404 Not Found (Book not found)
DELETE <code>/api/v1/books/{id}</code>	204 No Content	404 Not Found (Book not found)

5. API Versioning Convention

This API uses **URL Path Versioning**.

The version number is included in the URL path.

Example:

`/api/v1/books`

Using versioning allows future changes to be introduced without breaking existing clients.

6. Tools Used

- ASP.NET Core Web API
 - VS
 - Postman
 - Notion
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7. API Testing

The following endpoints were tested using Postman:

- Get all books
- Get book by ID
- Create a new book
- Update book information
- Delete a book
- Nested resource endpoint
- Error handling with 404 Not Found cases